

“AI-Powered Humanoid Robot Fleet Management System”

Team: Robo-Nexus

Team Members:

Kiran Khatri	kkhatri@sfsu.edu
Kerry Yu	kyu19@sfsu.edu
Dias Almat	dalmat@sfsu.edu
Ishaank Zalpuri	izalpuri@sfsu.edu
Etienne Ghashehbaba	eghashehbaba@mail.sfsu.edu

GitHub Repository:

<https://github.com/sfsu-joseo/csc675-775-database-api-project-robo-nexus>

Version	Description
V1	initial submission for review

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Project Description

This project focuses on designing a database system to manage fleets of humanoid robots used in environments such as warehouses, hospitals, and homes. As humanoid robots become more common, they generate a large amount of information, including task assignments, task execution history, deployment locations, energy levels, maintenance records, and interaction details. Currently, much of this information is either spread across different tools or not stored in a structured way, making it difficult to track robot behavior and performance over time.

There are existing robot fleet management platforms, such as `Formant` and `InOrbit`, that help monitor robots and show their current status. These systems are useful for real-time monitoring and coordination, but they are mainly designed for general robot fleets and do not focus on organizing long-term data for humanoid robots. In particular, they do not emphasize storing detailed interaction context, coordination roles, or historical data in a centralized database.

One unique feature of the system is its ability to store *human emotional context data* in personal and healthcare environments. In settings such as hospitals or homes, humanoid robots often interact closely with people. The system allows simple emotional observations, such as calm or urgent situations, to be recorded as data for later analysis. Most existing fleet management tools do not capture or organize this type of information.

Another unique feature is *role-based robot coordination* in warehouse environments. The system allows certain robots to be assigned a *senior* role to help coordinate tasks among other robots based on factors such as hardware capability and energy level. If a senior robot becomes unavailable, the system supports reassigning that role to another robot, helping prevent disruptions. Existing tools typically treat all robots equally and do not support this type of structured coordination with a fail-safe mechanism.

Overall, this project provides a centralized and structured way to manage humanoid robot data that existing solutions do not fully address. Manufacturers, warehouses, hospitals, and households deploying humanoid robots can benefit from a system that keeps robot information organized, reliable, and easy to understand over time.

Functional Database Requirements

1. Robot (Strong)

- 1.1. A robot shall belong at most one robot model.
- 1.2. A robot shall be deployed at at most one deployment location at a time.
- 1.3. A robot shall execute many tasks over time.
- 1.4. A robot shall be assigned at most one role at a time.
- 1.5. A robot shall have many emotion records over time.
- 1.6. A robot shall have at most one operator.
- 1.7. A robot shall have many energy status records over time.
- 1.8. A robot shall generate many logs.
- 1.9. A robot may be associated with many RobotHardware.
- 1.10. A robot shall have at most one software.
- 1.11. A robot shall have many performance metrics.
- 1.12. A robot shall generate many alerts.
- 1.13. A robot shall have many diagnostic reports over time.
- 1.14. A robot shall have many sensors
- 1.15. A robot shall charge at many stations
- 1.16. A robot shall have many charging sessions
- 1.17. A robot shall have a current status
- 1.18. A robot shall have an unique id

2. Robot Model (Strong)

- 2.1. A robot model shall be produced by at most one manufacturer.
- 2.2. A robot model shall be associated with many robots.
- 2.3. A robot model shall store the hardware and capability constraints of robots.

3. Hardware(Strong)

- 3.1. A hardware component shall be associated with many robots.
- 3.2. A hardware component shall be produced by at most one manufacturer.
- 3.3. A hardware component shall have a unique identifier.

4. Robot Hardware(Weak)

- 4.1. A robot hardware shall belong to exactly one robot.
- 4.2. A robot hardware shall belong to exactly one hardware.
- 4.3. A robot hardware shall record installation details.

5. **Manufacturer (Strong)**

- 5.1. A manufacturer shall produce many robot models.
- 5.2. A manufacturer shall have a unique identifier.
- 5.3. A manufacturer shall produce many hardware.

6. **Software Version (Strong)**

- 6.1. A software version shall be associated with many robots
- 6.2. A software version shall have an unique version number

7. **Deployment Location (Strong)**

- 7.1. A deployment location shall host many robots.
- 7.2. A deployment location shall support different environment types (warehouse, hospital, household).
- 7.3. A deployment location shall host many operators
- 7.4. A deployment location shall have many logs
- 7.5. A deployment location shall have many charging sessions

8. **Task (Strong)**

- 8.1. A task shall be defined independently of any robot.
- 8.2. A task shall be assigned to many robots.
- 8.3. A task shall have a priority level.

9. **Task Assignment (weak)**

- 9.1. A task assignment shall associate exactly one robot with one task.
- 9.2. A task assignment shall record the time at which a task was assigned.
- 9.3. A task assignment shall support reassignment of tasks over time.

10. **Task Execution (Weak)**

- 10.1. A task execution shall record the execution of one task by one robot.
- 10.2. A task execution shall have a start time and end time.
- 10.3. A task execution shall generate at least one log

11. Operator (User) (Strong)

- 11.1. An operator shall interact with many robots.
- 11.2. An operator shall be assigned many deployment locations.

12. Emotion Record (Weak)

- 12.1. An EmotionRecord shall store the observed emotional state associated with a human–robot interaction.
- 12.2. An EmotionRecord shall be associated with exactly one robot.
- 12.3. An EmotionRecord shall record the time at which the observation was made.
- 12.4. An EmotionRecord shall be associated with one deployment location where the interaction occurred.

13. Robot Role (Strong)

- 13.1. A robot role shall be assigned to many robots.
- 13.2. A robot role shall represent a coordination role such as senior or worker.

14. Energy Status (Weak)

- 14.1. An energy status shall be associated with exactly one robot.
- 14.2. An energy status shall record the battery level of the robot.

15. PerformanceMetrics(Weak)

- 15.1. A performance metrics shall belong to exactly one robot.
- 15.2. A performance shall have a time stamp record.

16. Alert(Weak)

- 16.1. An alert shall belong to exactly one robot.
- 16.2. An alert shall record severity and time.
- 16.3. An alert shall trigger AI request.

17. DiagnosticReport(Weak)

- 17.1. A diagnostic report shall belong to exactly one robot.
- 17.2. A diagnostic report shall have performance metrics.
- 17.3. A diagnostic report shall be analyzed by one AI model.

18. Log (weak)

- 18.1. A log shall be associated with one robot.
- 18.2. A log shall record operational events.
- 18.3. A log shall be associated with at most one deployment location

19. Sensor (Weak)

- 19.1. A sensor shall be associated with one robot
- 19.2. A sensor shall generate at least one log

20. Charging Station (Strong)

- 20.1. A charging station shall serve many robots
- 20.2. A charging station shall record many charging sessions

21. Charging Session (Weak)

- 21.1. A charging session shall occur at one charging station
- 21.2. A charging session shall be associated with one robot
- 21.3. A charging session shall be associated with one deployment location
- 21.4. A charging session shall generate at least one log

22. Support Request (Weak)

- 22.1. A support request shall be associated with at least one robot
- 22.2. A support request shall be generated by one operator
- 22.3. A support request shall generate at least one log
- 22.4. A support request shall record issues with associated robot

23. AI Model (Strong)

- 23.1. An AI model shall be associated with many robot models.
- 23.2. An AI model shall generate many AI requests.
- 23.3. An AI model shall store the type of services.

24. AI Request(Weak)

- 24.1. An AI request shall belong to exactly one robot.
- 24.2. An AI request shall associate with at least one AI Model.
- 24.3. An AI request shall have at least one diagnostic report.

25. AI Response(Weak)

- 25.1. An AI response shall belong to exactly one AI Model.
- 25.2. An AI response shall belong to exactly one robot.
- 25.3. An AI response shall store recommendation details.

26. Hospital(Strong)

- 26.1. A hospital shall have many robots
- 26.2. A hospital shall have many operators
- 26.3. A hospital shall have many charging station
- 26.4. A hospital shall belong to at most one deployment location
- 26.5. A hospital shall have many emotional records.
- 26.6. A hospital shall have many departments.

27. Warehouse(Strong)

- 27.1. A warehouse shall have many robots
- 27.2. A warehouse shall have many operators
- 27.3. A warehouse shall have many charging station
- 27.4. A warehouse shall belong to at most one deployment location

28. Home(Strong)

- 28.1. A home shall have many robots
- 28.2. A home shall have many operators
- 28.3. A home shall have many charging station
- 28.4. A home shall belong to at most one deployment location

29. Department(not sure)

- 29.1. A department shall belong to at most one hospital.
- 29.2. A department shall have many robots.
- 29.3. A department shall treat many patients.

Non-Functional Database Requirements

1. Performance

- 1.1. The system shall support concurrent access by at least _ robots without performance degradation.
- 1.2. The system shall be able to handle concurrent usage by _ robots
- 1.3. The system shall be able to maintain performance while receiving AI driven queries for real time data

2. Security

- 2.1. Any logs and data shall only be able to be accessed by an user with sufficient permissions
- 2.2. The system shall protect AI request and AI response data from unauthorized access.

3. Reliability

- 3.1. The system shall retain operational logs for at least 12 months.
- 3.2. The system shall have daily backups

4. Maintainability

- 4.1. The system shall host AI queries and structural queries separately
- 4.2. The database schema shall be well documented.

5. Scalability

- 5.1. The system shall support growth in the number of robots and deployment locations.
- 5.2. The system shall support increasing volumes of task execution and logs data.
- 5.3. The system shall support any future AI integration.

6. Privacy

- 6.1. Emotional interaction records shall be stored securely to protect human-related sensitive data.
- 6.2. Access to emotional data shall be restricted to authorized persons only .